

例 16-1 表 16-1 是一个研究吸烟、饮酒与食管癌关系的病例 - 对照资料, 试作 logistic 回归分析。

表 16-1 吸烟与食管癌关系的病例 - 对照调查资料

| 分层 $k$ | 吸烟 $X_1$ | 饮酒 $X_2$ | 观察例数 $n_k$ | 阳性数 $d_k$ | 阴性数 $n_k - d_k$ |
|--------|----------|----------|------------|-----------|-----------------|
| 1      | 0        | 0        | 199        | 63        | 136             |
| 2      | 0        | 1        | 170        | 63        | 107             |
| 3      | 1        | 0        | 101        | 44        | 57              |
| 4      | 1        | 1        | 416        | 265       | 151             |

例 16-2 为了探讨冠心病发生的有关危险因素, 对 26 例冠心病患者和 28 例对照者进行病例 - 对照研究, 各因素的说明及资料见表 16-2 和表 16-3, 试用 logistic 逐步回归分析方法筛选危险因素 ( $\alpha_{\lambda}=0.10, \alpha_{\text{III}}=0.15$ )。

表 16-2 冠心病 8 个可能的危险因素与赋值

| 因素        | 变量名   | 赋值说明                                  |
|-----------|-------|---------------------------------------|
| 年龄(岁)     | $X_1$ | $<45=1, 45\sim=2, 55\sim=3, 65\sim=4$ |
| 高血压史      | $X_2$ | 无=0, 有=1                              |
| 高血压家族史    | $X_3$ | 无=0, 有=1                              |
| 吸烟        | $X_4$ | 不吸=0, 吸=1                             |
| 高血脂史      | $X_5$ | 无=0, 有=1                              |
| 动物脂肪摄入    | $X_6$ | 低=0, 高=1                              |
| 体重指数(BMI) | $X_7$ | $<24=1, 24\sim=2, 26\sim=3$           |
| A 型性格     | $X_8$ | 否=0, 是=1                              |
| 冠心病       | $Y$   | 对照=0, 病例=1                            |

表 16-3 冠心病危险因素的病例 - 对照调查资料

| 序号 | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ | $Y$ |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-----|
| 1  | 3     | 1     | 0     | 1     | 0     | 0     | 1     | 1     | 0   |
| 2  | 2     | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0   |
| 3  | 2     | 1     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 4  | 2     | 0     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 5  | 3     | 0     | 0     | 1     | 0     | 1     | 1     | 1     | 0   |
| 6  | 3     | 0     | 1     | 1     | 0     | 0     | 2     | 1     | 0   |
| 7  | 2     | 0     | 1     | 0     | 0     | 0     | 1     | 0     | 0   |
| 8  | 3     | 0     | 1     | 1     | 1     | 0     | 1     | 0     | 0   |
| 9  | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0   |
| 10 | 1     | 0     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 11 | 1     | 0     | 1     | 0     | 0     | 0     | 1     | 1     | 0   |
| 12 | 1     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0   |
| 13 | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0   |
| 14 | 4     | 1     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 15 | 3     | 0     | 1     | 1     | 0     | 0     | 1     | 1     | 0   |
| 16 | 1     | 0     | 0     | 1     | 0     | 0     | 3     | 1     | 0   |

续表

| 序号 | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ | $Y$ |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-----|
| 17 | 2     | 0     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 18 | 1     | 0     | 0     | 1     | 0     | 0     | 1     | 1     | 0   |
| 19 | 3     | 1     | 1     | 1     | 1     | 0     | 1     | 0     | 0   |
| 20 | 2     | 1     | 1     | 1     | 1     | 0     | 2     | 0     | 0   |
| 21 | 3     | 1     | 0     | 1     | 0     | 0     | 1     | 0     | 0   |
| 22 | 2     | 1     | 1     | 0     | 1     | 0     | 3     | 1     | 0   |
| 23 | 2     | 0     | 0     | 1     | 1     | 0     | 1     | 1     | 0   |
| 24 | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0   |
| 25 | 2     | 0     | 1     | 0     | 0     | 0     | 1     | 0     | 0   |
| 26 | 2     | 0     | 0     | 1     | 1     | 0     | 1     | 1     | 0   |
| 27 | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0   |
| 28 | 2     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0   |
| 29 | 2     | 1     | 1     | 1     | 0     | 1     | 2     | 1     | 1   |
| 30 | 3     | 0     | 0     | 1     | 1     | 1     | 2     | 1     | 1   |
| 31 | 2     | 0     | 0     | 1     | 1     | 1     | 1     | 0     | 1   |
| 32 | 3     | 1     | 1     | 1     | 1     | 1     | 3     | 1     | 1   |
| 33 | 2     | 0     | 0     | 1     | 0     | 0     | 1     | 1     | 1   |
| 34 | 2     | 0     | 1     | 0     | 1     | 1     | 1     | 1     | 1   |
| 35 | 2     | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1   |
| 36 | 2     | 1     | 1     | 1     | 1     | 0     | 1     | 1     | 1   |
| 37 | 3     | 1     | 1     | 1     | 1     | 0     | 1     | 1     | 1   |
| 38 | 3     | 1     | 1     | 1     | 0     | 1     | 1     | 1     | 1   |
| 39 | 3     | 1     | 1     | 1     | 1     | 0     | 1     | 1     | 1   |
| 40 | 3     | 0     | 1     | 0     | 0     | 0     | 1     | 0     | 1   |
| 41 | 2     | 1     | 1     | 1     | 1     | 0     | 2     | 1     | 1   |
| 42 | 3     | 1     | 0     | 1     | 0     | 1     | 2     | 1     | 1   |
| 43 | 3     | 1     | 0     | 1     | 0     | 0     | 1     | 1     | 1   |
| 44 | 3     | 1     | 1     | 1     | 1     | 1     | 2     | 0     | 1   |
| 45 | 4     | 0     | 0     | 1     | 1     | 0     | 3     | 1     | 1   |
| 46 | 3     | 1     | 1     | 1     | 1     | 0     | 3     | 1     | 1   |
| 47 | 4     | 1     | 1     | 1     | 1     | 0     | 3     | 0     | 1   |
| 48 | 3     | 0     | 1     | 1     | 1     | 0     | 1     | 1     | 1   |
| 49 | 4     | 0     | 0     | 1     | 0     | 0     | 2     | 1     | 1   |
| 50 | 1     | 0     | 1     | 1     | 1     | 0     | 2     | 1     | 1   |
| 51 | 2     | 0     | 1     | 1     | 0     | 1     | 2     | 1     | 1   |
| 52 | 2     | 1     | 1     | 1     | 0     | 0     | 2     | 1     | 1   |
| 53 | 2     | 1     | 0     | 1     | 0     | 0     | 1     | 1     | 1   |
| 54 | 3     | 1     | 1     | 0     | 1     | 0     | 3     | 1     | 1   |